Univariate Linear Regression

General Principles

To study relationships between a continuous independent variable and a continuous dependent variable (e.g., height and weight), we can use linear regression. Essentially, we draw a line that passes through the point cloud of the two variables being tested. For this, we need to have:

1) An intercept α, which represents the origin of the line,i.e., the expected value of the dependent variable (height) when the independent variable (weight) is equal to zero.
2) A coefficient β, which informs us about the slope of the line. In other words, it tells us how much Y (height) increases for each increment of the independent variable (weight).
3) A standard deviation term σ, which informs us about the spread of points around the line, i.e., the variance around the prediction.

Considerations

> **Note**
>
> - Bayesian models allow us to update our understanding of parameters conditional on an observed data set. This allows us to consider model parameter uncertainty , which quantifies our confidence or uncertainty in the parameters in the form of a posterior distribution . Therefore, we need to declare prior distributions for each model parameter, in this case for: α, β, and σ.
> - Prior distributions are built following these considerations:
>   - As the data are normalized (see introduction), we can use a Normal distribution for α and β, with a mean of 0 and a standard deviation of 1. This tends to be a weakly regularizing prior, and weaker priors like a $Normal(0, 10)$ are also possible.
>   - Since σ must be strictly positive, we must use a distribution with support on the positive reals, such as the *Exponential* or *Folded-Normal* distribution.

- Gaussian regression deals directly with continuous outcomes, estimating a linear relationship between predictors and the outcome variable without depending on a non linear link function (see introduction). This simplifies interpretation, as coefficients represent direct changes in the outcome variable.

Example

Below is an example code snippet demonstrating *Bayesian linear regression* using the BayesForge (**BF**) package. Data consist of two continuous variables (height and weight), and the goal is to estimate the effect of weight on height. This example is based on McElreath (2018).

Python

```python
from BayesForge import bf

# Setup device-------------------------------------------------
m = bf(platform="cpu")

# Import Data & Data Manipulation ------------------------------------------------
# Import
data_path = m.load.howell1(only_path=True)
m.data(data_path, sep=";")
m.df = m.df[m.df.age > 18]  # Subset data to adults
m.scale(["weight"])  # Normalize


# Define model -------------------------------------------------
def model(weight, height):
    a = m.dist.normal(178, 20, name="a")
    b = m.dist.log_normal(0, 1, name="b")
    s = m.dist.uniform(0, 50, name="s")
    m.dist.normal(a + b * weight, s, obs=height)


# Run mcmc -------------------------------------------------
m.fit(model, progress_bar=False)  # Optimize model parameters through MCMC sampling

# Summary -------------------------------------------------
m.summary()  # Get posterior distributions
```

```
/home/sosa/.local/lib/python3.13/site-packages/tqdm/auto.py:21: TqdmWarning:

IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs

An NVIDIA GPU may be present on this machine, but a CUDA-enabled jaxlib is not installed. Fal

bf v 0.0.60 package loaded
jax.local_device_count 32
```

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	mcse_mean	mcse_sd
a	154.65	0.27	154.22	155.08	3707.02	2798.78	1.0	0.0	0.0
b	5.82	0.28	5.37	6.27	3821.10	2728.88	1.0	0.0	0.0
s	5.14	0.20	4.83	5.47	3531.18	2719.49	1.0	0.0	0.0

R

```r
library(BayesForge)
m <- importBF(platform = "cpu")

# Load csv file
m$data(m$load$howell1(only_path = T), sep = ";")

# Filter data frame
m$df <- m$df[m$df$age > 18, ] # Subset data to adults

# Scale
m$scale(list("weight")) # Normalize

# Convert data to JAX arrays
m$data_to_model(list("weight", "height"))

# Define model ------------------------------------------------
model <- function(height, weight) {
    # Parameter prior distributions
    s <- bf.dist.uniform(0, 50, name = "s")
    a <- bf.dist.normal(178, 20, name = "a")
    b <- bf.dist.normal(0, 1, name = "b")

    # Likelihood
```

```
    bf.dist.normal(a + b * weight, s, obs = height)
}

# Run MCMC ------------------------------------------------
m$fit(model) # Optimize model parameters through MCMC sampling

# Summary -------------------------------------------------
m$summary()
```

Julia

```
using PyBayesForge

# Setup device--------------------------------------------------
m = importBF(platform="cpu")

# Import Data & Data Manipulation ------------------------------------------------
# Import
data_path = m.load.howell1(only_path = true)
m.data(data_path, sep=';')
m.df = m.df[m.df.age > 18] # Subset data to adults
m.scale(["weight"]) # Normalize

# Define model -------------------------------------------------
@BF function model(weight, height)
    # Priors
    a = m.dist.normal(178, 20, name = 'a')
    b = m.dist.log_normal(0, 1, name = 'b')
    s = m.dist.uniform(0, 50, name = 's')
    m.dist.normal(a + b * weight , s, obs = height)
end

# Run mcmc -----------------------------------------------
m.fit(model)  # Optimize model parameters through MCMC sampling

# Summary -------------------------------------------------
m.summary() # Get posterior distributions
```

Mathematical Details

Frequentist Formulation

The following equation describe the frequentist formulation of linear regression:

$$Y_i = \alpha + \beta X_i + \epsilon_i$$

Where:

- Y_i is the dependent variable for observation i.
- α is the intercept term.
- β is the regression coefficient.
- X_i is the input variable for observation i.
- ϵ_i is the error term for observation i, and the vector of the error terms, ϵ, are assumed to be independent and identically distributed.

Bayesian Formulation

In the Bayesian formulation, we define each parameter with priors . We can express a Bayesian version of this regression model using the following model:

$$Y_i \sim \text{Normal}(\alpha + \beta X_i, \sigma)$$

$$\alpha \sim \text{Normal}(0, 1)$$

$$\beta \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{Uniform}(0, 50)$$

Where:

- Y_i is the dependent variable for observation i.
- α and β are the intercept and regression coefficient, respectively.
- X_i is the independent variable for observation i.

- σ is the standard deviation of the Normal distribution, which describes the variance in the relationship between the dependent variable Y and the independent variable X.

Model validation tests

- Comparison with Stan: `1.Continuous variable.py` fits the same model with BayesForge and Stan and compares the posterior distributions (KL divergence).
- Documentation example: the Python example of this page is run by the documentation tests (`test_model_pages.py`).

Notes

> **Note**
>
> We observe a difference between the *Frequentist* and the *Bayesian* formulation regarding the error term. Indeed, in the *Frequentist* formulation, the error term ϵ represents residual fluctuations around the predicted values. This assumption leads to point estimates for α and β. In contrast, the *Bayesian* formulation treats σ as a parameter with its own prior distribution. This allows us to incorporate our uncertainty about the error term into the model.

Reference(s)

McElreath, Richard. 2018. *Statistical Rethinking: A Bayesian course with examples in R and Stan*. Chapman; Hall/CRC.